

HR Analytics - Storyboard & Data Model

Assessment 5: Telling the Story in Power BI

Data Model - Star Schema

The flat HR_Analytics table (1,470 rows, 19 columns) was split into a star schema using the Python script in src/build_star_schema.py. All five relationships run one-to-many from each dimension to the central Fact_Employee table, with a single cross-filter direction.

Table	Type	Primary key	Rows	Relationship to Fact
Fact_Employee	FACT - central	EmpID	1,470	Central table
Dim_Department	Dimension	DepartmentID	3	DepartmentID FK - 1:many
Dim_Education	Dimension	EducationFieldID	6	EducationFieldID FK - 1:many
Dim_JobRole	Dimension	JobRoleID	9	JobRoleID FK - 1:many
Dim_AgeGroup	Dimension	AgeGroupID	5	AgeGroupID FK - 1:many
Dim_SalarySlab	Dimension	SalarySlabID	4	SalarySlabID FK - 1:many

All 5 relationships: One-to-many (1:*) from Dimension to Fact_Employee. Cross-filter direction = Single. Set in Power BI Model View.

Report Page Flow - Three Pages

Page	Purpose	Key visuals & confirmed values	Decision
Page 1 Executive Summary	Single-glance workforce overview	5 KPI cards: 1,470 employees · 237 attritions · 16.1% rate · Age 38.9 · R7,676 avg salary Donut: Life Sciences 38.4% · Bar: Upto 5k = 129 attritions · Column: 26-35 = 83	Is attrition a priority, and where is the biggest risk?
Page 2 Department Drill-Down	Role, satisfaction, gender and tenure analysis	Bar: Lab Technician 59, Sales Exec 54, Research Scientist 44 Matrix: Role × Satisfaction showing varying counts Treemap: Male 142, Female 95 · Area: peaks at year 0-2	Which roles and tenure bands are at the highest risk?
Page 3 Tenure & Salary	Salary and tenure vs attrition	Scatter: Income vs tenure by attrition status. Bar: Sales R7,877 · R&D; R7,719 · HR R7,040. Cards: Avg 5.53 years · 1,470 employees	Which salary bands need urgent intervention?

Submission Files

File	Folder	Status
Fact_Employee.csv + 5 Dim_*.csv	data/star_schema/	✓ Done
HR_Analytics.csv (original flat file)	data/raw/	✓ Done
build_star_schema.py	src/	✓ Done
star_schema.sql	sql/	✓ Done
HR_Analytics_Report.pbix	src/	✓ Done
Storyboard.pdf	docs/	✓ Done
Narrative.pdf	docs/	✓ Done
Accessibility_Checklist.pdf	docs/	✓ Done
Prompt_Reflection_Log.pdf	docs/	✓ Done
README.md	root/	✓ Done